

MINSI LU

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EDUCATION BACKGROUND

Department of Pharmacy science, Tsinghua University	Beijing, China
Bachelor of Science in Pharmaceutical Sciences	Aug. 2019 - Jul. 2024
Paul G. Allen School of Computer Science and Engineering, University of Washington	Seattle, US
Visiting Student	Jun. 2023 - Oct. 2023
Department of Electrical & Computer Engineering, University of Waterloo	Waterloo, Canada
Master of Engineering in Electrical & Computer Engineering	Sep. 2025 - Present

PUBLICATIONS

Bowen Gao, Bo Qiang, Haichuan Tan, Minsi Ren, Yinjun Jia, **Minsi Lu**, Jingjing Liu, Weiyang Ma, Yanyan Lan, 'DrugCLIP: Contrastive Protein-Molecule Representation Learning for Virtual Screening', **NeurIPS 2023**.

Tangqi Fang, Yifeng Liu, Addie Woicik, **Minsi Lu**, Anupama Jha, Xiao Wang, Gang Li, Borislav Hristov, Zixuan Liu, Hanwen Xu, William S. Noble, Sheng Wang, 'Enhancing Hi-C Contact Matrices for Loop Detection with Capricorn', **ISMB 2024**.

RESEARCH EXPERIENCE

Research Assistant, School of Computer Science & Engineering, University of Washington	Seattle, US
Project: Resolution Enhancement and Time Dimension Modeling for Hi-C Data	Jun. 2023 - Oct. 2023
<i>Advisor: Prof. William Stafford Noble, Prof. Sheng Wang</i>	
- Aimed to integrate the time and other feature dimensions of Hi-C data for resolution enhancement. - Developed Capricorn, a tool for Hi-C resolution enhancement that incorporates high-order chromatin features as additional views of the input Hi-C contact matrix and leverages a diffusion probability model backbone to generate a high-resolution matrix. - Formed a related paper that was submitted to ISMB. The link is https://academic.oup.com/bioinformatics/article/40/Supplement_1/i471/7700867	
Research Assistant, Institute for AI Industry Research, Tsinghua University	Beijing, China
Project: Contrastive Learning for Drug Virtual Screening	Feb. 2023 - Jun. 2023
<i>Advisor: Prof. YanYan Lan</i>	
- Aimed to develop a new method for virtual screening, which is a critical step in AI-assisted drug discovery. - Proposed a novel contrastive learning framework, DrugCLIP, by reformulating virtual screening as a dense retrieval task and employing contrastive learning to align representations of binding protein pockets and molecules from a large quantity of pairwise data without explicit binding-affinity scores. - Formed a related paper to introduce the novel feature representation methodology, was submitted to NeurIPS. The link is https://neurips.cc/virtual/2023/poster/70609	

PROJECTS

University of Waterloo	Waterloo, Canada
Project: Multimodal Proteomics Modeling for peptide-spectrum alignment	Jan. 2025 - Jun. 2025
<i>Advisor: Prof. Ming Li</i>	
- Redesigned and retrained a 197M-parameter CoCa-style multimodal encoder-decoder model for spectrum-peptide alignment, extending DeepSearch with a 16-layer spectrum Transformer encoder and PTM-aware multimodal peptide decoder. - Jointly optimized in-batch contrastive learning (cross-modal retrieval) and autoregressive de novo sequencing objectives, enabling unified spectrum-peptide embedding space and sequence generation under teacher-forcing.	
Project: WildChat-MCP: Hierarchical Analysis Framework via Model Context Protocol	Sep. 2025 - Dec. 2025
<i>Advisor: Prof. Jimmy Lin</i>	
- Implemented a Model Context Protocol (MCP) server enabling LLMs (Claude Desktop) to autonomously explore and analyze a 1.4M+ conversation dataset via structured tools and resources. - Built a full ETL pipeline (HuggingFace streaming → DuckDB → hybrid topic modeling) including	

embedding-based clustering and LLM-based cluster labeling (Qwen 7B). Optimized large-scale updates using CTAS and indexed the database with B-Tree + BM25 FTS for low-latency retrieval.

- Architected a hierarchical agent tool stack ("Meta → Macro → Discovery → Micro") with safety-sandboxed SQL execution.
- Repo: <https://github.com/minsilu/WildChat-MCP>

**Project: Biology-Informed Filtering Adaptive ANN Search
for Tandem Mass Spectrometry**

Sep. 2025 - Dec. 2025

Advisor: Prof. Xiao Hu

- Designed a hybrid range-filtered approximate nearest neighbor (RF-ANNS) system for tandem MS open search, integrating mass-aware columnar storage with HNSW-based graph search to address large-tolerance (± 500 Da) peptide identification
- Proposed Rescue-HNSW, a mass-aware dual-neighbor graph architecture with adaptive pre/post-filtering based on query selectivity, mitigating graph connectivity loss under aggressive filtering.
- Achieved 13 $\times$ speedup over MSFragger (500 Da tolerance) while maintaining competitive identification rates at 1% FDR

Tsinghua University

Beijing, China

Project: Multiomics Integration Via Graph Learning

Sep. 2022 - Jan. 2023

Advisor: Prof. Jianyang Zeng

- Aimed to integrate scRNA-seq data and scATAC-seq data through graph learning based on GLUE.
- Independently decayed the weight of edges according to the distance of the ATAC peak from the TSS and used a set of independent aggregators to combine the messages in each GCN layer.
- Our method achieved comparative performance with GLUE on metrics, and had the ability to identify more genes with regulatory relationships than GLUE.
- Repo: <https://github.com/minsilu/Multiomics-Integration-via-Graph-Learning>

Project: Gomoku Genius: AI Search Strategies for Classic Board Game Mastery

Feb. 2023 - Jun. 2023

Advisor: Prof. Mingsheng Long

- Aimed to investigate and implement five distinct AI agents for board games: Minimax Search, Alpha-Beta Pruning, Truncated Search, Monte Carlo Tree Search (MCTS), and Evaluation Function-based AlphaZero, addressing the challenges of search complexity and function design.
- Independently: Optimized Minimax for Tic-Tac-Toe. Applied Alpha-Beta Pruning to reduce search space. Implemented Truncated Search in Go-Moku, balancing time constraints with strategic depth using pattern evaluation. Conducted gameplay analysis comparing naive MCTS and Alpha-Beta. Enhanced MCTS with evaluation function, significantly improving board state assessments and decision-making.
- A practical application of advanced AI search strategies, demonstrating superior performance in board games.
- Repo: <https://github.com/minsilu/Gomoku-Genius--AI-Search-Strategies-for-Classic-Board-Game-Mastery>

SKILLS

Mathematics: Calculus, Linear Algebra, Probability and Statistics.

Programming Skills: Skilled at C/C++, Java, Python, R, SQL. Familiar with Linux/Unix.

Machine Learning & AI: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-Learn; Experienced in Deep Learning architecture design and deployment.

Systems & Infrastructure: Linux/Unix, Git, Docker, Kubernetes; Cloud Computing (AWS) and CI/CD pipelines.

AWARDS & HONORS

Scholarship for Excellent in Academic, University Scholarship (10%)	2023
First Prize in 16 th "Novozymes Cup" Life Science Culture Festival, Municipal Award (10%)	2021
Third Prize in 39 th "Challenges Cup" Student Academic Competition, University Award	2021
Scholarship for Progress in Academic, University Scholarship	2021
Third Prize in Winter Vacation Social Practice for Students, University Award	2021
Gold Medal in 27 th China Biology Olympiad(CBO), National Award	2018

SOCIETY ROLES

Leader at Student Winter Vacation Social Practice Team.

Dec. 2020 - Jan. 2021